

SaaS Lead Generation & Revenue Intelligence Pipeline

BILL SWIFT

An end-to-end automation project for Bill Swift, a usage-based billing SaaS company

A portfolio project by Praveen Sambasivam

What is Bill Swift?

Bill Swift is a fictional SaaS company built as the business scenario for this portfolio project — modeled on Chargebee, a real subscription billing and revenue management platform used by SaaS companies. Bill Swift gives every part of this pipeline a reason to exist: a webinar, a scored lead, a CRM record, and a revenue forecast that a real billing-SaaS company's sales and finance teams would actually use.

What is this project about?

This project is an automated, end-to-end SaaS lead-generation and revenue-intelligence pipeline. From the moment someone registers for a webinar to the moment their deal closes, every step — scoring, CRM sync, email nurture, revenue prediction, and reporting — runs without manual intervention.

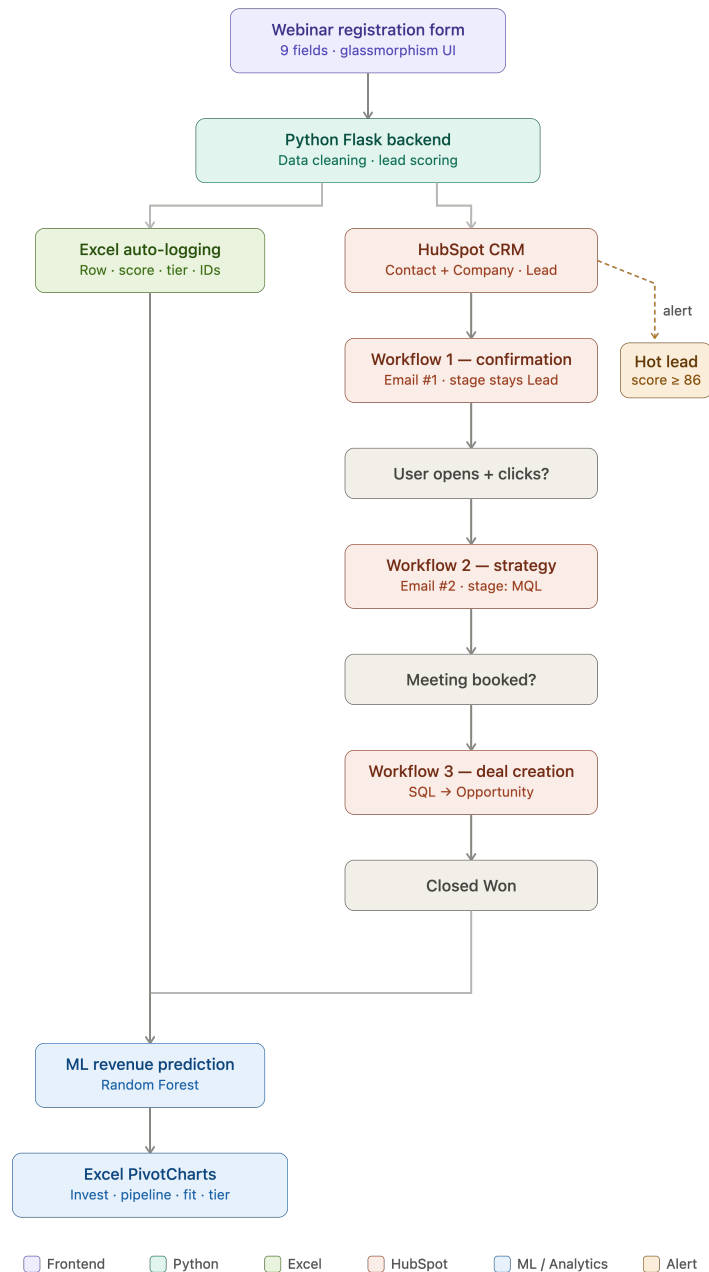
What problem does this project solve?

Every SaaS company faces the same gap between a new lead and a closed deal — leads come in faster than anyone can manually qualify, track, or follow up with. For Bill Swift, I built a system to close that entire gap automatically:

- **Lead qualification** — a 6-signal, 100-point scoring engine scores every lead automatically based on the details captured at signup (company size, transaction volume, current billing setup, and more), so sales isn't guessing who to prioritize.
- **CRM and data consistency** — every lead syncs to Excel and HubSpot simultaneously with matching IDs, keeping one reliable source of truth instead of two drifting records.
- **Hot lead response time** — an automated alert flags any lead scoring above 85, so high-value prospects aren't missed while sitting in a queue.
- **Structured follow-up** — tiered nurture workflows move leads through Lead → MQL → SQL → Opportunity → Closed Won without manual handoffs.
- **Revenue visibility** — an ML revenue-prediction model feeds CXO-ready PivotCharts, showing leadership where the pipeline is strongest and where to invest.

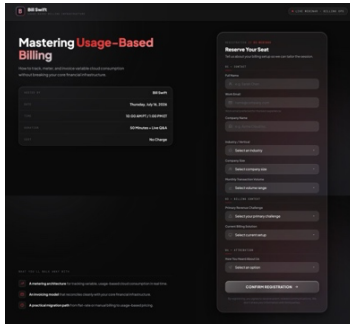
Pipeline Overview

Here's how a lead moves through the system, end to end:

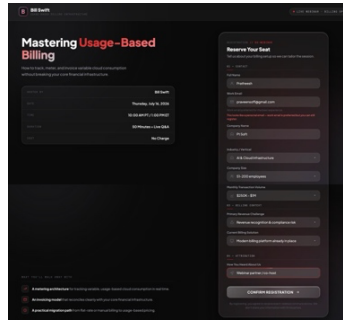


1. Webinar registration form

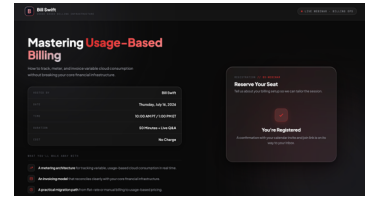
Lead-gen entry point — a webinar signup form that captures contact, company, and billing context in one step.



Full webpage — event details and registration form



Form mid-fill — personal email soft warning shown



Success screen — confirms the pipeline is live

2. Python Flask backend

Receives the form data, cleans it, and triggers scoring — all in one request.

3. Lead scoring engine

Scores each lead out of 100 across 6 signals; sorts into Hot / Warm / Cold.

Signal	Max
Monthly Transaction Volume	28
Primary Revenue Challenge	24
Current Billing Solution	19
Company Size	14
How You Heard About Us	10
Email Type	5

Tiers: 0-35 Cold · 36-65 Warm · 66-100 Hot

Six weighted scoring signals feed a 100-point scale, split into three tiers.

4. Excel auto-logging

Auto-appends every lead to Excel with score, tier, and IDs — no manual entry.

Timestamp	Name	Email	Company	Personal	Industry	Industry	Company Size	Monthly Revenue	Primary Revenue Challenge	Current Billing Solution	How You Heard About Us	Lead Score	Lead Tier	HubSpot ID	HubSpot ID	Predict
2026-07-01	Praveen Sai	praveen.dur	Prvn Tech	Personal	Er edtech		201-1000	250k-1m	manual-inv	stripe-payp	google	58	Warm	51244505	53152383	152837.3
2026-07-01	Sarah Chen	sarah.sarah	veri	Vertex AI	vertexai.io	ecommerce	201-1000	50k-250k	usage-mete	in-house	newsletter	65	Warm	51248541	53157054	227258
2026-07-01	David Ross	dross@saa	Saasflow	saasflow.cc	proptech	iot	201-1000	250k-1m	usage-mete	in-house	partner	75	Hot	51244342	53153833	271457.6
2026-07-01	Elena Rost	elena@cyb	Cybercore	cybercore.ai	fintech		11-50	10m+	research	stripe-payp	google	60	Warm	51245502	53149321	184046.1
2026-07-01	Elena Rost	elena@cyb	Cybercore	cybercore.ai	ai-cloud		11-50	250k-1m	research	legacy-ente	referral	45	Warm	51245502	53149321	126484.1
2026-07-01	Marcus Var	marcus@p	Paybolt	paybolt.fin	ecommerce		51-200	1m-10m	research	legacy-ente	newsletter	51	Warm	51238017	53154755	157615.1
2026-07-01	Chloe Tana	c.tanaka@	Edulearn	edulearn.ci	edtech		1-10	under-50k	rev-recogni	in-house	partner	47	Warm	51235872	53147174	180323.1
2026-07-01	Liam O'Con	liam@stres	Streamvib	streamvibe	digital-media		1-10	250k-1m	usage-mete	none-man	other	66	Hot	51242147	53154585	238010.9
2026-07-01	Amara Okal	a.okafor@	Nexthome	nexthome.bi	proptech	iot	51-200	10m+	research	stripe-payp	linkedin	65	Warm	51235194	53153292	193846.6
2026-07-01	Jacob Mille	j.miller@bi	Biotrack	biotrack.or	ai-cloud		1000+	50k-250k	manual-inv	none-man	google	62	Warm	51244507	53149501	171892.2
2026-07-01	Pratheesh	praveensof	Pt Soft	Personal Er	ai-cloud		51-200	250k-1m	rev-recogni	modern-pl	partner	49	Warm	51238751	53152383	189583.2

Every submission appends a new row — score, tier, and HubSpot IDs included.

5. HubSpot CRM sync

Creates Contact + Company in HubSpot; handles duplicate conflicts automatically.

Name	Email	Phone Number	Contact owner	Primary comp
Praveen Sambasivam	praveens1499@gmail.com	909 909 9090	Praveen Sambasivam	Praveen
Praveen Sambasivam	praveens1499@gmail.com	909 909 9090	Praveen Sambasivam	Praveen
Praveen Sambasivam	praveens1499@gmail.com	909 909 9090	Praveen Sambasivam	Praveen
Praveen Sambasivam	praveens1499@gmail.com	909 909 9090	Praveen Sambasivam	Praveen
Praveen Sambasivam	praveens1499@gmail.com	909 909 9090	Praveen Sambasivam	Praveen
Praveen Sambasivam	praveens1499@gmail.com	909 909 9090	Praveen Sambasivam	Praveen
Praveen Sambasivam	praveens1499@gmail.com	909 909 9090	Praveen Sambasivam	Praveen
Praveen Sambasivam	praveens1499@gmail.com	909 909 9090	Praveen Sambasivam	Praveen
Praveen Sambasivam	praveens1499@gmail.com	909 909 9090	Praveen Sambasivam	Praveen
Praveen Sambasivam	praveens1499@gmail.com	909 909 9090	Praveen Sambasivam	Praveen

HubSpot Contacts — auto-created on every submission

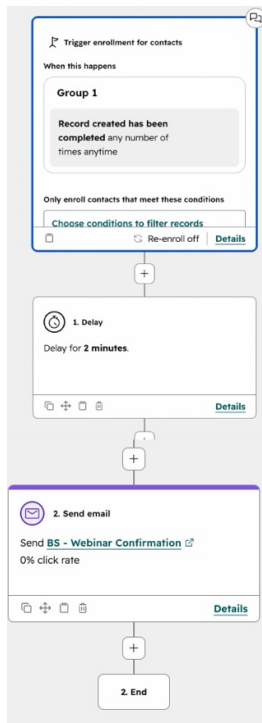
Company name	Company owner	Create date	Phone Number	Last Activity
Praveen Sambasivam	No owner	Today at 1:00 AM GMT+0:00	---	---
Praveen Sambasivam	No owner	Today at 1:00 AM GMT+0:00	---	---
Praveen Sambasivam	No owner	Today at 1:00 AM GMT+0:00	---	---
Praveen Sambasivam	No owner	Today at 1:00 AM GMT+0:00	---	---
Praveen Sambasivam	No owner	Today at 1:00 AM GMT+0:00	---	---
Praveen Sambasivam	No owner	Today at 1:00 AM GMT+0:00	---	---
Praveen Sambasivam	No owner	Today at 1:00 AM GMT+0:00	---	---
Praveen Sambasivam	No owner	Today at 1:00 AM GMT+0:00	---	---
Praveen Sambasivam	No owner	Today at 1:00 AM GMT+0:00	---	---
Praveen Sambasivam	No owner	Today at 1:00 AM GMT+0:00	---	---

HubSpot Companies — linked to matching contacts

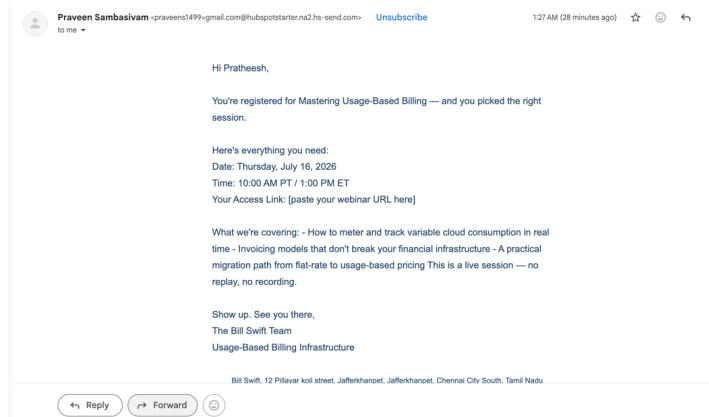
6. Automated workflows

4 workflows: confirmation email → MQL upgrade → SQL/deal creation → hot lead alert.

Workflow 1 — registration confirmation

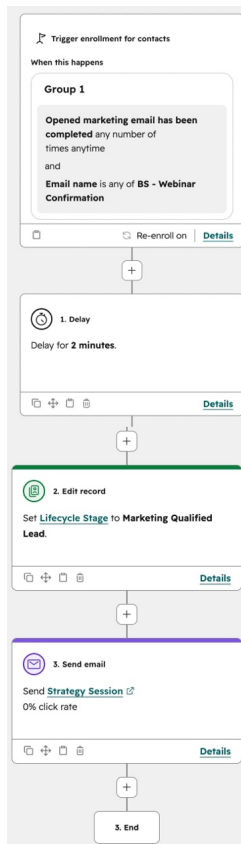


Fires on contact creation, sends confirmation email

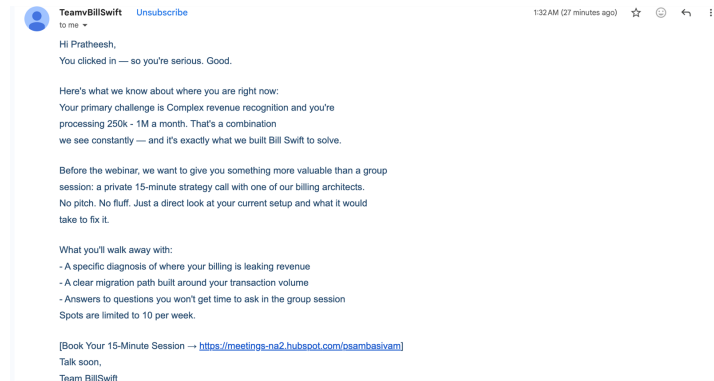


Email #1 received — personalized with webinar details

Workflow 2 — MQL upgrade & strategy session

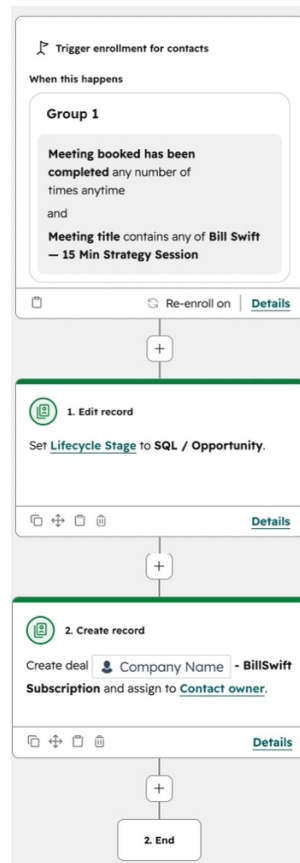


Email open triggers MQL upgrade, sends Email #2



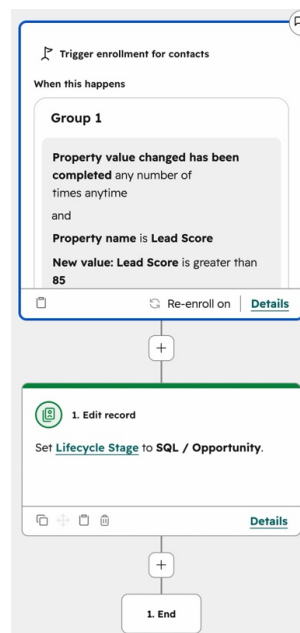
Email #2 — dynamic tokens populate real challenge + volume

Workflow 3 — meeting booked → SQL & deal creation



Meeting booked triggers SQL/Opportunity and auto-creates a deal.

Workflow 4 — hot lead alert (Lead Score > 85)



Lead Score above 85 triggers an instant SQL upgrade and sales alert — independent of the email sequence.

7. Deal creation & Closed Won

Meeting booked → deal auto-created in pipeline → progresses to Closed Won.

The screenshot displays a competition management interface with a horizontal timeline of stages. The stages are: 'Discover a Challenge' (active), 'Product Demo / Proposal', 'Contract Review', 'Contract Negotiation', 'Closed Won', and 'Closed Lost'. The 'Discover a Challenge' stage is expanded, showing details for a challenge titled 'PI Soft - B2BSoft' with a subscription. The challenge was created on 07/02/2026 and is owned by Praveen Sambasivam. A dropdown menu is open below the owner's name, showing 'H...' and 'B...' options. A note indicates 'Bidding in 7 days'.

Deal auto-created at Discovery Scheduled — no manual entry into the pipeline.

8. ML revenue prediction

Random Forest model predicts contract value; written back to Excel.

Project ID	Project Name	Start	Current Status	Category	Industry	Priority	Health	Complexity	Team Size	Progress (%)	Key Milestones	Next Steps	Lead	Team Lead	Project Manager	Overall Status	Annual Review
2023-01-001	Alpha Core	2023-01-01	Completed	Software	IT	High	Green	Low	5	100	Phase 1: Planning	Phase 2: Development	John Doe	John Doe	John Doe	Completed	2023-01-01
2023-01-002	Beta Launch	2023-01-05	In Progress	Marketing	Marketing	Medium	Yellow	Medium	3	75	Phase 3: Testing	Phase 4: Deployment	Jane Smith	Jane Smith	Jane Smith	In Progress	2023-01-05
2023-01-003	Gamma Upgrade	2023-01-10	On Hold	Hardware	Hardware	Low	Red	High	8	20	Phase 5: Review	Phase 6: Post-mortem	Mike Johnson	Mike Johnson	Mike Johnson	On Hold	2023-01-10
2023-01-004	Delta Integration	2023-01-15	Completed	Software	IT	High	Green	Low	4	100	Phase 7: Planning	Phase 8: Development	Sarah Brown	Sarah Brown	Sarah Brown	Completed	2023-01-15
2023-01-005	Epsilon Rollout	2023-01-20	In Progress	Marketing	Marketing	Medium	Yellow	Medium	3	60	Phase 9: Testing	Phase 10: Deployment	David Wilson	David Wilson	David Wilson	In Progress	2023-01-20
2023-01-006	Zeta Upgrade	2023-01-25	On Hold	Hardware	Hardware	Low	Red	High	6	30	Phase 11: Review	Phase 12: Post-mortem	Emily Davis	Emily Davis	Emily Davis	On Hold	2023-01-25
2023-01-007	Eta Integration	2023-02-01	Completed	Software	IT	High	Green	Low	5	100	Phase 13: Planning	Phase 14: Development	Chris Evans	Chris Evans	Chris Evans	Completed	2023-02-01
2023-01-008	Theta Launch	2023-02-05	In Progress	Marketing	Marketing	Medium	Yellow	Medium	4	80	Phase 15: Testing	Phase 16: Deployment	Alexander Lee	Alexander Lee	Alexander Lee	In Progress	2023-02-05
2023-01-009	Iota Upgrade	2023-02-10	On Hold	Hardware	Hardware	Low	Red	High	7	40	Phase 17: Review	Phase 18: Post-mortem	Olivia White	Olivia White	Olivia White	On Hold	2023-02-10
2023-01-010	Kappa Integration	2023-02-15	Completed	Software	IT	High	Green	Low	6	100	Phase 19: Planning	Phase 20: Development	Benjamin Green	Benjamin Green	Benjamin Green	Completed	2023-02-15
2023-01-011	Lambda Launch	2023-02-20	In Progress	Marketing	Marketing	Medium	Yellow	Medium	5	90	Phase 21: Testing	Phase 22: Deployment	Mia Black	Mia Black	Mia Black	In Progress	2023-02-20
2023-01-012	Mu Upgrade	2023-02-25	On Hold	Hardware	Hardware	Low	Red	High	4	50	Phase 23: Review	Phase 24: Post-mortem	Noah Brown	Noah Brown	Noah Brown	On Hold	2023-02-25
2023-01-013	Nu Integration	2023-03-01	Completed	Software	IT	High	Green	Low	7	100	Phase 25: Planning	Phase 26: Development	Ava White	Ava White	Ava White	Completed	2023-03-01
2023-01-014	Xi Launch	2023-03-05	In Progress	Marketing	Marketing	Medium	Yellow	Medium	6	85	Phase 27: Testing	Phase 28: Deployment	Liam Black	Liam Black	Liam Black	In Progress	2023-03-05
2023-01-015	Omicron Upgrade	2023-03-10	On Hold	Hardware	Hardware	Low	Red	High	5	60	Phase 29: Review	Phase 30: Post-mortem	Emma Brown	Emma Brown	Emma Brown	On Hold	2023-03-10
2023-01-016	Pi Integration	2023-03-15	Completed	Software	IT	High	Green	Low	8	100	Phase 31: Planning	Phase 32: Development	Lucas White	Lucas White	Lucas White	Completed	2023-03-15
2023-01-017	Rho Launch	2023-03-20	In Progress	Marketing	Marketing	Medium	Yellow	Medium	7	95	Phase 33: Testing	Phase 34: Deployment	Sophia Black	Sophia Black	Sophia Black	In Progress	2023-03-20
2023-01-018	Sigma Upgrade	2023-03-25	On Hold	Hardware	Hardware	Low	Red	High	6	70	Phase 35: Review	Phase 36: Post-mortem	Isabella Brown	Isabella Brown	Isabella Brown	On Hold	2023-03-25
2023-01-019	Tau Integration	2023-04-01	Completed	Software	IT	High	Green	Low	9	100	Phase 37: Planning	Phase 38: Development	Jack White	Jack White	Jack White	Completed	2023-04-01
2023-01-020	Upsilon Launch	2023-04-05	In Progress	Marketing	Marketing	Medium	Yellow	Medium	8	88	Phase 39: Testing	Phase 40: Deployment	Chloe Black	Chloe Black	Chloe Black	In Progress	2023-04-05
2023-01-021	Phi Upgrade	2023-04-10	On Hold	Hardware	Hardware	Low	Red	High	7	55	Phase 41: Review	Phase 42: Post-mortem	Leo Brown	Leo Brown	Leo Brown	On Hold	2023-04-10
2023-01-022	Chi Integration	2023-04-15	Completed	Software	IT	High	Green	Low	10	100	Phase 43: Planning	Phase 44: Development	Aria White	Aria White	Aria White	Completed	2023-04-15
2023-01-023	Psi Launch	2023-04-20	In Progress	Marketing	Marketing	Medium	Yellow	Medium	9	92	Phase 45: Testing	Phase 46: Deployment	Ethan Black	Ethan Black	Ethan Black	In Progress	2023-04-20
2023-01-024	Omega Upgrade	2023-04-25	On Hold	Hardware	Hardware	Low	Red	High	8	65	Phase 47: Review	Phase 48: Post-mortem	Ava Brown	Ava Brown	Ava Brown	On Hold	2023-04-25
2023-01-025	Alpha Integration	2023-05-01	Completed	Software	IT	High	Green	Low	11	100	Phase 49: Planning	Phase 50: Development	Noah White	Noah White	Noah White	Completed	2023-05-01
2023-01-026	Beta Launch	2023-05-05	In Progress	Marketing	Marketing	Medium	Yellow	Medium	10	90	Phase 51: Testing	Phase 52: Deployment	Mia Black	Mia Black	Mia Black	In Progress	2023-05-05
2023-01-027	Gamma Upgrade	2023-05-10	On Hold	Hardware	Hardware	Low	Red	High	9	45	Phase 53: Review	Phase 54: Post-mortem	Leo Brown	Leo Brown	Leo Brown	On Hold	2023-05-10
2023-01-028	Delta Integration	2023-05-15	Completed	Software	IT	High	Green	Low	12	100	Phase 55: Planning	Phase 56: Development	Aria White	Aria White	Aria White	Completed	2023-05-15
2023-01-029	Epsilon Rollout	2023-05-20	In Progress	Marketing	Marketing	Medium	Yellow	Medium	11	85	Phase 57: Testing	Phase 58: Deployment	Ethan Black	Ethan Black	Ethan Black	In Progress	2023-05-20
2023-01-030	Zeta Upgrade	2023-05-25	On Hold	Hardware	Hardware	Low	Red	High	10	50	Phase 59: Review	Phase 60: Post-mortem	Ava Brown	Ava Brown	Ava Brown	On Hold	2023-05-25
2023-01-031	Eta Integration	2023-06-01	Completed	Software	IT	High	Green	Low	13	100	Phase 61: Planning	Phase 62: Development	Noah White	Noah White	Noah White	Completed	2023-06-01
2023-01-032	Theta Launch	2023-06-05	In Progress	Marketing	Marketing	Medium	Yellow	Medium	12	95	Phase 63: Testing	Phase 64: Deployment	Mia Black	Mia Black	Mia Black	In Progress	2023-06-05
2023-01-033	Iota Upgrade	2023-06-10	On Hold	Hardware	Hardware	Low	Red	High	11	60	Phase 65: Review	Phase 66: Post-mortem	Leo Brown	Leo Brown	Leo Brown	On Hold	2023-06-10
2023-01-034	Kappa Integration	2023-06-15	Completed	Software	IT	High	Green	Low	14	100	Phase 67: Planning	Phase 68: Development	Aria White	Aria White	Aria White	Completed	2023-06-15
2023-01-035	Lambda Launch	2023-06-20	In Progress	Marketing	Marketing	Medium	Yellow	Medium	13	90	Phase 69: Testing	Phase 70: Deployment	Ethan Black	Ethan Black	Ethan Black	In Progress	2023-06-20
2023-01-036	Mu Upgrade	2023-06-25	On Hold	Hardware	Hardware	Low	Red	High	12	55	Phase 71: Review	Phase 72: Post-mortem	Ava Brown	Ava Brown	Ava Brown	On Hold	2023-06-25
2023-01-037	Nu Integration	2023-07-01	Completed	Software	IT	High	Green	Low	15	100	Phase 73: Planning	Phase 74: Development	Noah White	Noah White	Noah White	Completed	2023-07-01
2023-01-038	Xi Launch	2023-07-05	In Progress	Marketing	Marketing	Medium	Yellow	Medium	14	88	Phase 75: Testing	Phase 76: Deployment	Mia Black	Mia Black	Mia Black	In Progress	2023-07-05
2023-01-039	Omicron Upgrade	2023-07-10	On Hold	Hardware	Hardware	Low	Red	High	13	65	Phase 77: Review	Phase 78: Post-mortem	Leo Brown	Leo Brown	Leo Brown	On Hold	2023-07-10
2023-01-040	Pi Integration	2023-07-15	Completed	Software	IT	High	Green	Low	16	100	Phase 79: Planning	Phase 80: Development	Aria White	Aria White	Aria White	Completed	2023-07-15
2023-01-041	Rho Launch	2023-07-20	In Progress	Marketing	Marketing	Medium	Yellow	Medium	15	92	Phase 81: Testing	Phase 82: Deployment	Ethan Black	Ethan Black	Ethan Black	In Progress	2023-07-20
2023-01-042	Sigma Upgrade	2023-07-25	On Hold	Hardware	Hardware	Low	Red	High	14	70	Phase 83: Review	Phase 84: Post-mortem	Ava Brown	Ava Brown	Ava Brown	On Hold	2023-07-25
2023-01-043	Tau Integration	2023-08-01	Completed	Software	IT	High	Green	Low	17	100	Phase 85: Planning	Phase 86: Development	Noah White	Noah White	Noah White	Completed	2023-08-01
2023-01-044	Upsilon Launch	2023-08-05	In Progress	Marketing	Marketing	Medium	Yellow	Medium	16	95	Phase 87: Testing	Phase 88: Deployment	Mia Black	Mia Black	Mia Black	In Progress	2023-08-05
2023-01-045	Phi Upgrade	2023-08-10	On Hold	Hardware	Hardware	Low	Red	High	15	60	Phase 89: Review	Phase 90: Post-mortem	Leo Brown	Leo Brown	Leo Brown	On Hold	2023-08-10
2023-01-046	Chi Integration	2023-08-15	Completed	Software	IT	High	Green	Low	18	100	Phase 91: Planning	Phase 92: Development	Aria White	Aria White	Aria White	Completed	2023-08-15
2023-01-047	Psi Launch	2023-08-20	In Progress	Marketing	Marketing	Medium	Yellow	Medium	17	90	Phase 93: Testing	Phase 94: Deployment	Ethan Black	Ethan Black	Ethan Black	In Progress	2023-08-20
2023-01-048	Omega Upgrade	2023-08-25	On Hold	Hardware	Hardware	Low	Red	High	16	55	Phase 95: Review	Phase 96: Post-mortem	Ava Brown	Ava Brown	Ava Brown	On Hold	2023-08-25
2023-01-049	Alpha Integration	2023-09-01	Completed	Software	IT	High	Green	Low	19	100	Phase 97: Planning	Phase 98: Development	Noah White	Noah White	Noah White	Completed	2023-09-01
2023-01-050	Beta Launch	2023-09-05	In Progress	Marketing	Marketing	Medium	Yellow	Medium	18	95	Phase 99: Testing	Phase 100: Deployment	Mia Black	Mia Black	Mia Black	In Progress	2023-09-05

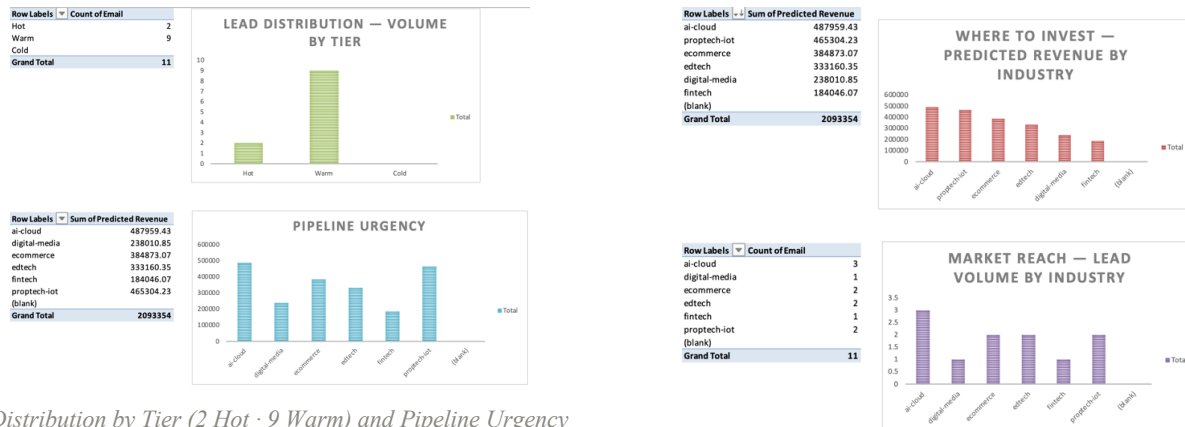
200-row historical dataset used to train the Random Forest model at server startup.

	Lead Score	Lead Time	HubSpot	HubSpot	Predicted
	58	Warm	51244505	33152383	152837.3
er	65	Warm	51248541	33157054	227258
	75	Hot	51244342	33153833	271457.6
	60	Warm	51245502	33149321	184046.1
	45	Warm	51245502	33149321	126484.1
er	51	Warm	51238017	33154755	157615.1
	47	Warm	51235872	33147174	180323.1
	66	Hot	51242147	33154585	238010.9
	65	Warm	51235194	33153292	193846.6
	62	Warm	51244507	33149501	171892.2
	49	Warm	51238751	33152383	189583.2

Predicted revenue is written back into the same Excel row as the lead score.

9. Excel PivotCharts

4 live charts: revenue by industry, lead volume, tier distribution, pipeline urgency.



Lead Distribution by Tier (2 Hot · 9 Warm) and Pipeline Urgency

Where to Invest — revenue by industry, and Market Reach by volume

Project Summary

- Zero manual steps — from form submission to CRM contact, ML prediction, and email delivery, everything is automated.
- Server-side lead scoring — scores are calculated in Python and cannot be tampered with from the browser.
- Dynamic email personalization — Email #2 injects real CRM properties into each send.
- Full audit trail — every row contains a timestamp, cleaned data, lead score, tier, HubSpot IDs, and predicted revenue.